

Assignment Specification

IoT Standards and Protocols

Context

- 100% of your overall mark!
 - Project ethos (1 overall project)
- For this assignment you are required to:
 - Propose a project.
 - Create a working project using networking/IoT standards & protocols.
 - Requires a "physical" aspect – sensors/devices(or simulated by request)
 - Present and communicate your work in a clear, correct manner.
- The project should incorporate the different layers (sensor, processing node, gateway, application) of an IoT/connected device architecture.
- Project will be assessed on its technical (e.g., features) content, complexity, applicability to domain and project communication/execution

Requirements

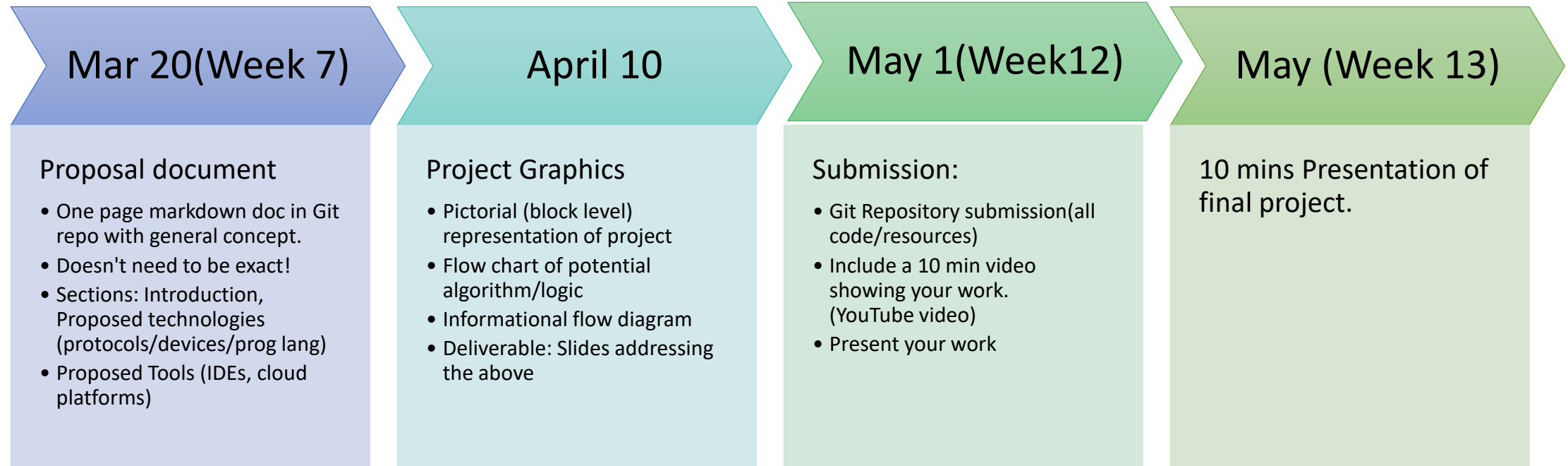
"Include the different layers (sensor, processing, gateway, application) in an IoT Project."

Design	Propose and design solutions (e.g. See attached work packages)
Apply	Apply suitable IoT protocols and standards.
Model/ implement	Model/implement a solution to your proposal Use the knowledge, skills and practices from other modules Should be scoped correctly – can't build a production standard solution in a few weeks!
Present/curate	Present/curate your project Create a short video that demonstrates the project.

IoT Layers of your Project

- **Sensor Layer:** Collects real-time data from the environment.
- **Processing Node:** Raspberry Pi processes and transmits data.
- **Gateway Layer:** Data is relayed through an IoT hub/platform.
- **Application Layer:** Visualises and actuates responses.

Timeline

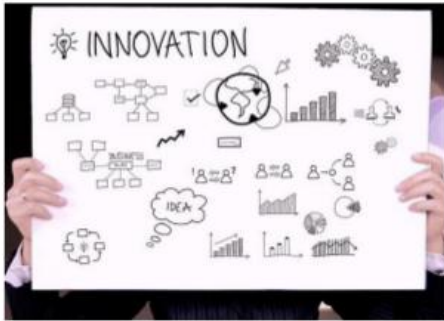


Assignment Requirements

Each person/team must **propose, design, and implement** an **IoT-based solution** that includes:

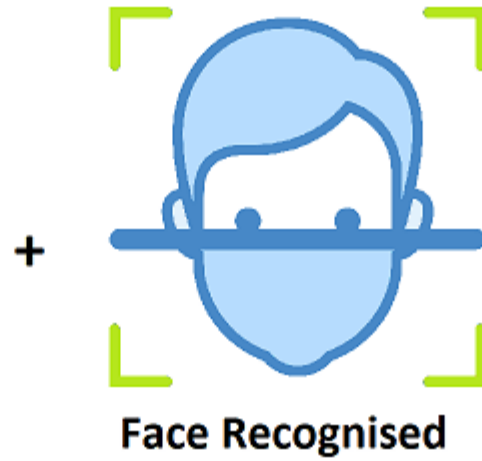
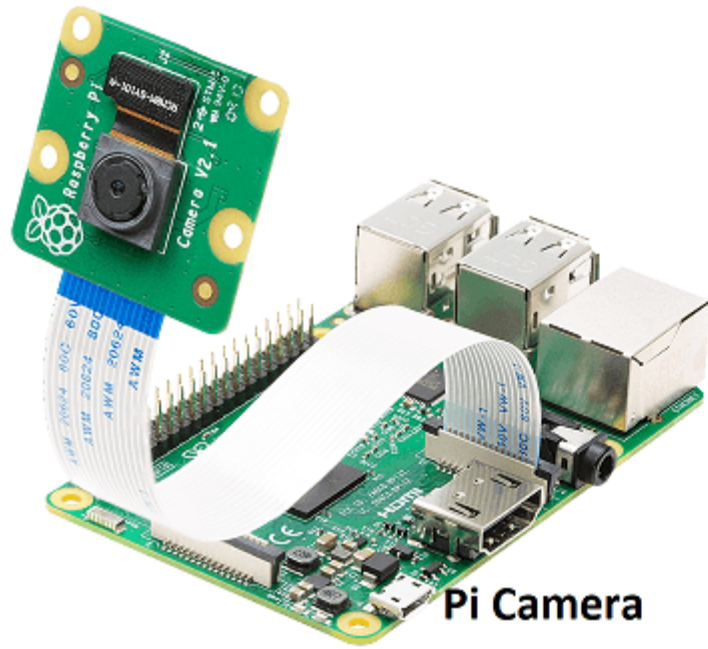
Requirement	Description	Weight
IoT Device Connection	- Set up a Raspberry Pi 4 or similar device connected to an IoT messaging platform or server . - Configure MQTT communication or HTTP-based messaging . - Verify data transmission from the device to the platform/server .	10%
Sensor Data Collection	- Choose at least two sensors for monitoring (e.g., temperature, motion, light, etc.). - Write a Python script to collect real-time data and store it locally. - Ensure the data is timestamped .	15%
Telemetry Transmission	- Format sensor readings as JSON . - Implement secure data transmission using MQTT or HTTP to the IoT server/platform. - Apply error handling and retry mechanisms .	15%
Data Visualisation	- Use a dashboard or visualisation tool (e.g., IoT dashboard/ web interface) to display sensor trends. - Display key parameters (e.g., Heart Rate, motion activity). - Ensure real-time updates are reflected on the dashboard.	15%
Data Processing & Actuation	- Implement an automated response based on sensor readings . - Use a server-side script or event-processing system to analyse incoming data and trigger actions (e.g., turning on a fan, sending an alert). - If applicable, integrate computer vision or AI-based analysis for real-time decision-making.	30%
Project Presentation & Documentation	Submit a report that includes: - System architecture diagrams . - Code snippets and explanations . - Screenshots of dashboards and telemetry data . - Challenges encountered and solutions implemented . - A 5-minute demonstration video showcasing the project.	15%

Other Considerations



- "Permissionless Innovation"!
 - You don't have to limit yourself to module technologies. You can integrate other devices/ systems
 - We used Python but you can use any other programming languages/ frameworks if you want. (Remember, the RPi can run Node/Java/JavaScript)
- Have a look at other IoT project examples for inspiration.
- It's OK to "simulate sensors"
 - You can simulate data (e.g. GPS location.)

Possible Concept: Home Monitoring/Ambient/Assisted Living



Device unlocked

ProjectGraphic

